

Name: Talha UmerStudent ID: 23201097 Sec:

1. (10 marks) Consider the following fixed point function, $f(x) = \frac{9x^3 - 7x^2 + 5x}{x}$. Show the first five iterations starting from $x=1$.
2. (5 marks) Show the derivation for the number of iterations needed for a bisection method.

$$f(x) = \frac{9x^3 - 7x^2 + 5x}{x}$$

$$9x^3 - 7x^2 + 5x = 0$$

$$5x = 7x^2 - 9x^3$$

$$x = \frac{7}{5}x^2 - \frac{9}{5}x^3$$

$$g(x) = \frac{7}{5}x^2 - \frac{9}{5}x^3$$

Starting from $x=1$.

$$g(1) = \frac{7}{5}(1)^2 - \frac{9}{5}(1)^3 = -\frac{2}{5}$$

$$g\left(-\frac{2}{5}\right) =$$

$$f(x) = \frac{9x^3 - 7x^2 + 5x}{x}$$

$$= 9x^2 - 7x + 5$$

$$\therefore g(1) = 9(1)^2 - 7(1) + 5$$

$$= 9 - 7 + 5 = 7$$

$$\therefore g(7) = 9(7)^2 - 7(7) + 5$$

$$= 397$$

$$\therefore g(397) = 9(397)^2 - 7(397) + 5$$

$$= 1415707$$

$$\therefore g(1415707) = 9(1415707)^2 - 7(1415707) + 5$$

$$= 1.8 \times 10^{13}$$

$$\therefore g(1.8 \times 10^{13}) = 9(1.8 \times 10^{13})^2 - 7(1.8 \times 10^{13}) + 5$$

$$= 2.923 \times 10^{27}$$

b)

$$|b_n - a_n| = \frac{|b_0 - a_0|}{2^n}$$

~~$|b_0 - a_0|$~~

$$|\text{Actual Error}| = \frac{b_k - a_k}{2}$$

$$|m_n - n^*| = \frac{|b_0 - a_0|}{2^n}$$

$$\frac{|b_0 - a_0|}{2^{k+1}} \leq \epsilon < \epsilon$$

$$\lg \left(\frac{|b_0 - a_0|}{2^{k+1}} \right) \leq \lg \epsilon$$

$$\frac{\lg(|b_0 - a_0|)}{\lg 2} - \lg 2^{k+1} \leq \lg \epsilon$$

$$\lg(|b_0 - a_0|) - \lg \epsilon \leq (k+1) \lg 2$$

$$k \geq \frac{\lg(|b_0 - a_0|) - \lg \epsilon}{\lg 2} + 1$$